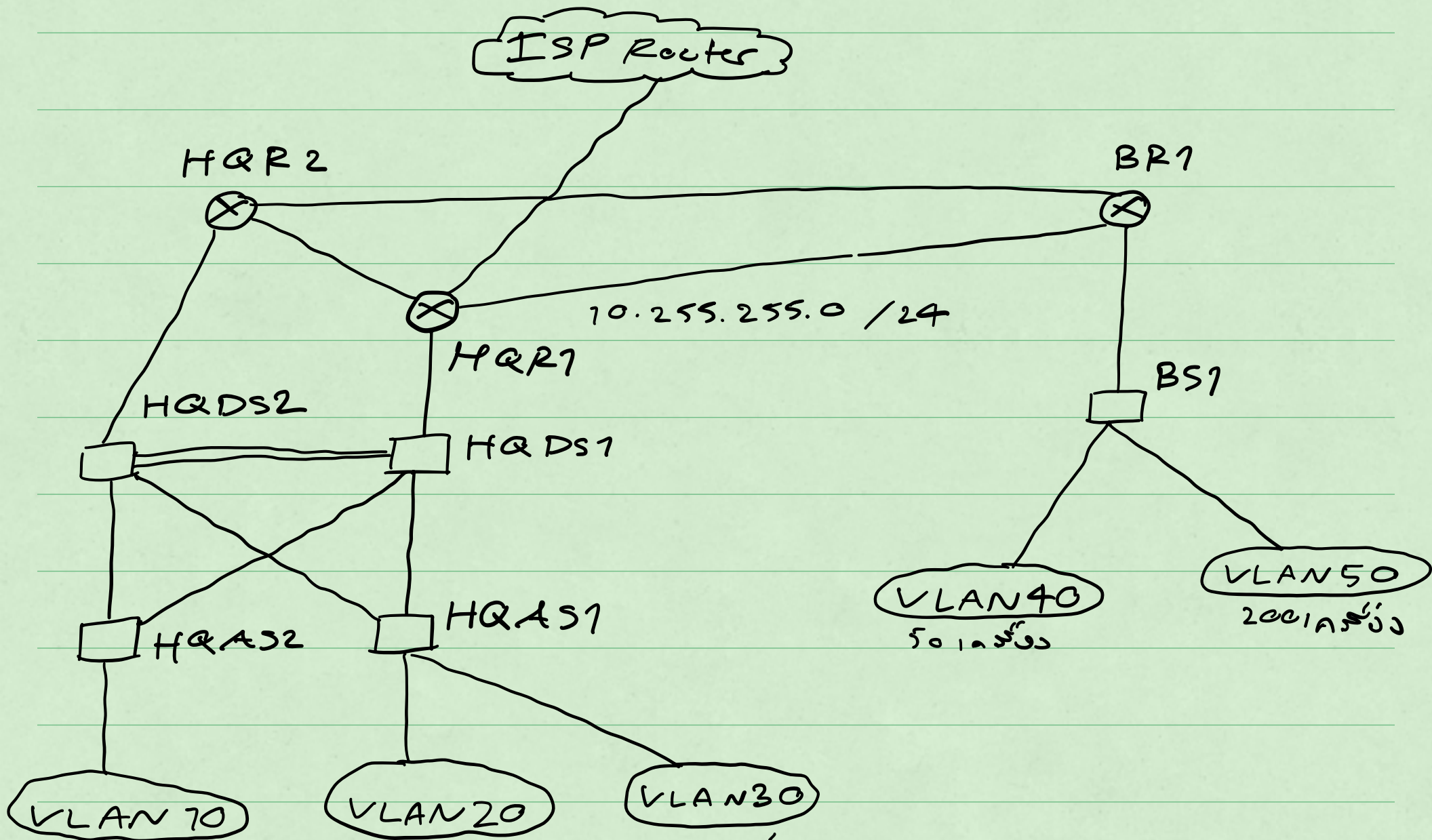




30 hosts

60 hosts

120 hosts

public IP
WAN - 203.0.113.0/30 , 10.255.255.0/24

private IP
LAN → 172.16.0.0/16

① Subnetting (VLSM) & IP convention

- Finding first usable address & 11010 = 22 subnet
အဲဒီ virtual router IP အားပေးတာ HSRP
- IP ကိုယ့်အတွက်ပဲ physical IP ပဲ router လေးကို
sub interface
- IP အားပေးတာပဲ အဲဒီမှာပဲ 11010 ကို host

for detailed calculation please check out →

1) 11010 = LAN - 172.16.0.0/16

VLAN 70 (IT & Admin) 30 လောက်ပဲ 0 = 7/2

network address အဲဒီ 172.16.2.0/27

broadcast address

first usable

last usable
virtual IP

VLAN 20 ()

},

2) 110000 WAN

public WAN - 203.0.113.0/30

first usable address @ 203.0.113.1/30

last usable address @ 203.0.113.2/30

private WAN - 10.255.255.0/24

need 3 subnet 110 = 11010 = subnet 952125

02702

subnet 1

first usable @

last usable @

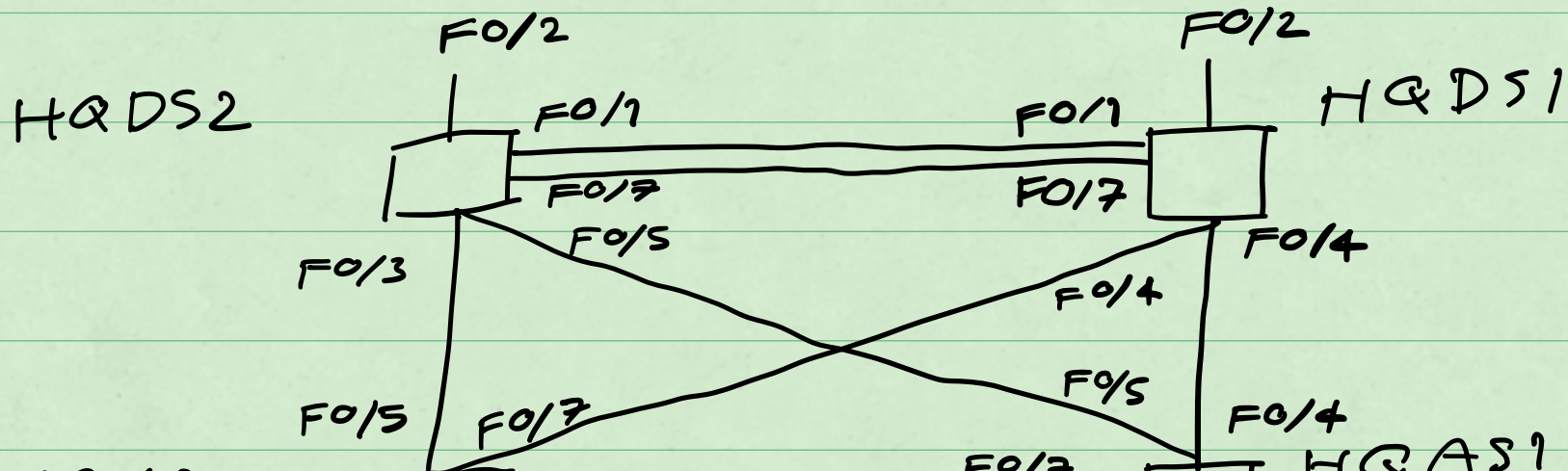
subnet 2

•
•
•

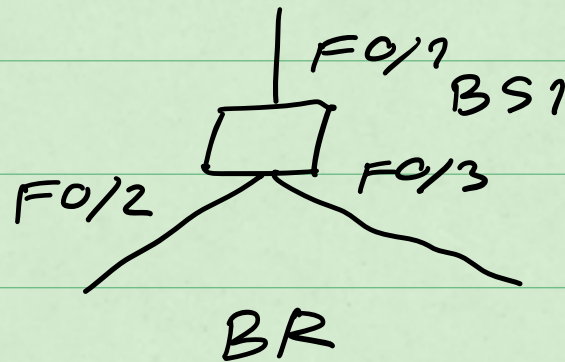
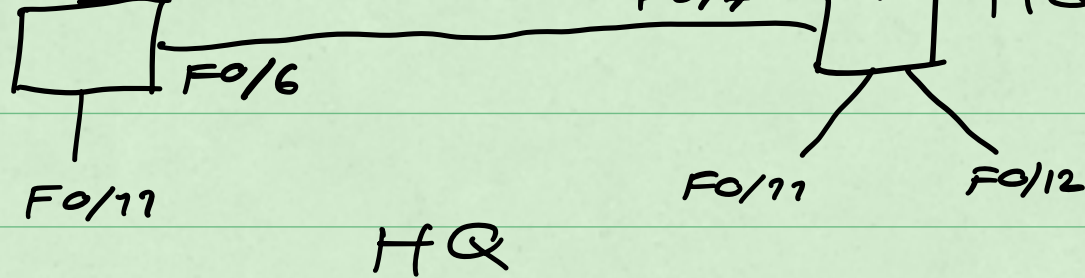
② Layer 2

Trunking

- Full interface configuration on host & PC
is trunk port & access port = full & access
access port

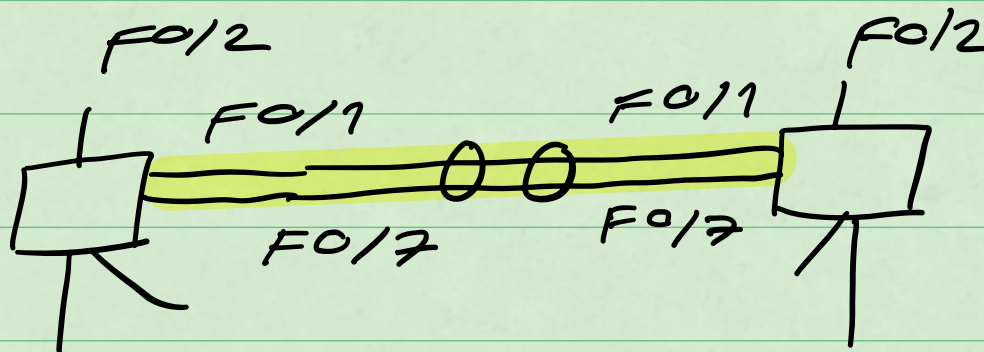


HQAS2

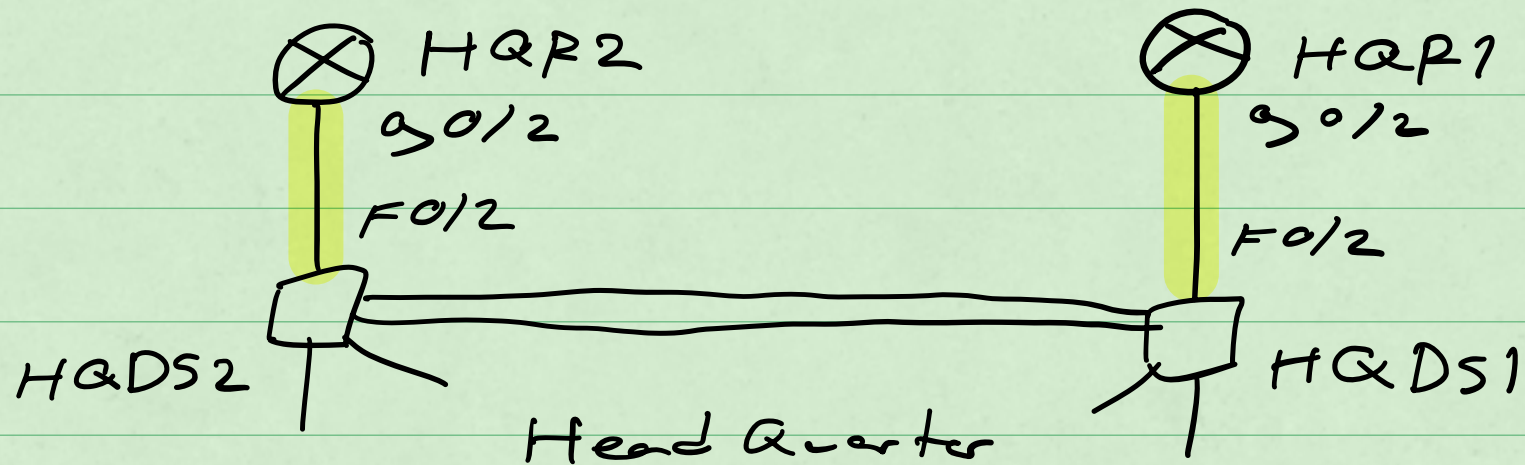


Ether Channel

ಇದರಲ್ಲಿ 5 = 40 Gbps HQDS2 110 = HQDS1 ನಿಗೆ Ether Chan
1100 Lacp ಡಾನ್ಡ್



ROAS



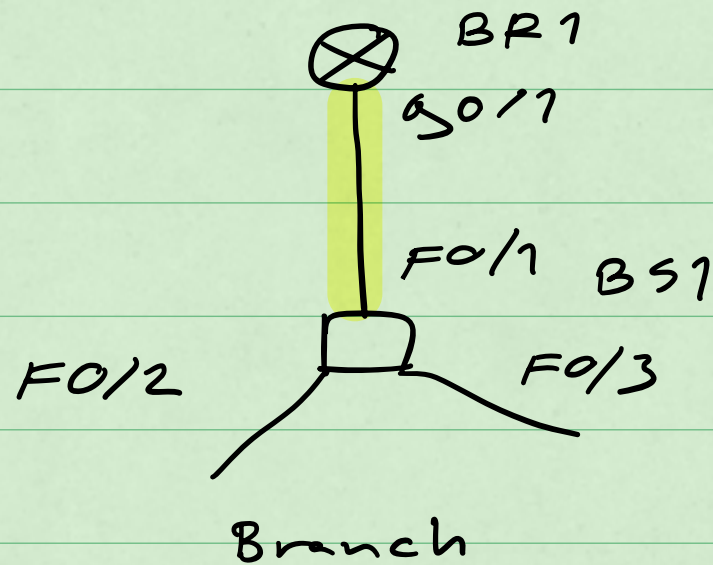
no router on a stick scenario HQ router ==
HQ Distributed Switch

HQR2 == HQR1 subinterface

go/2.10 == VLAN 10

go/2.20 == VLAN 20

go/2.30 == VLAN 30



in router on a stick setup BR1 == BS1

BR1 subinterface

go/1.40 serve VLAN 40

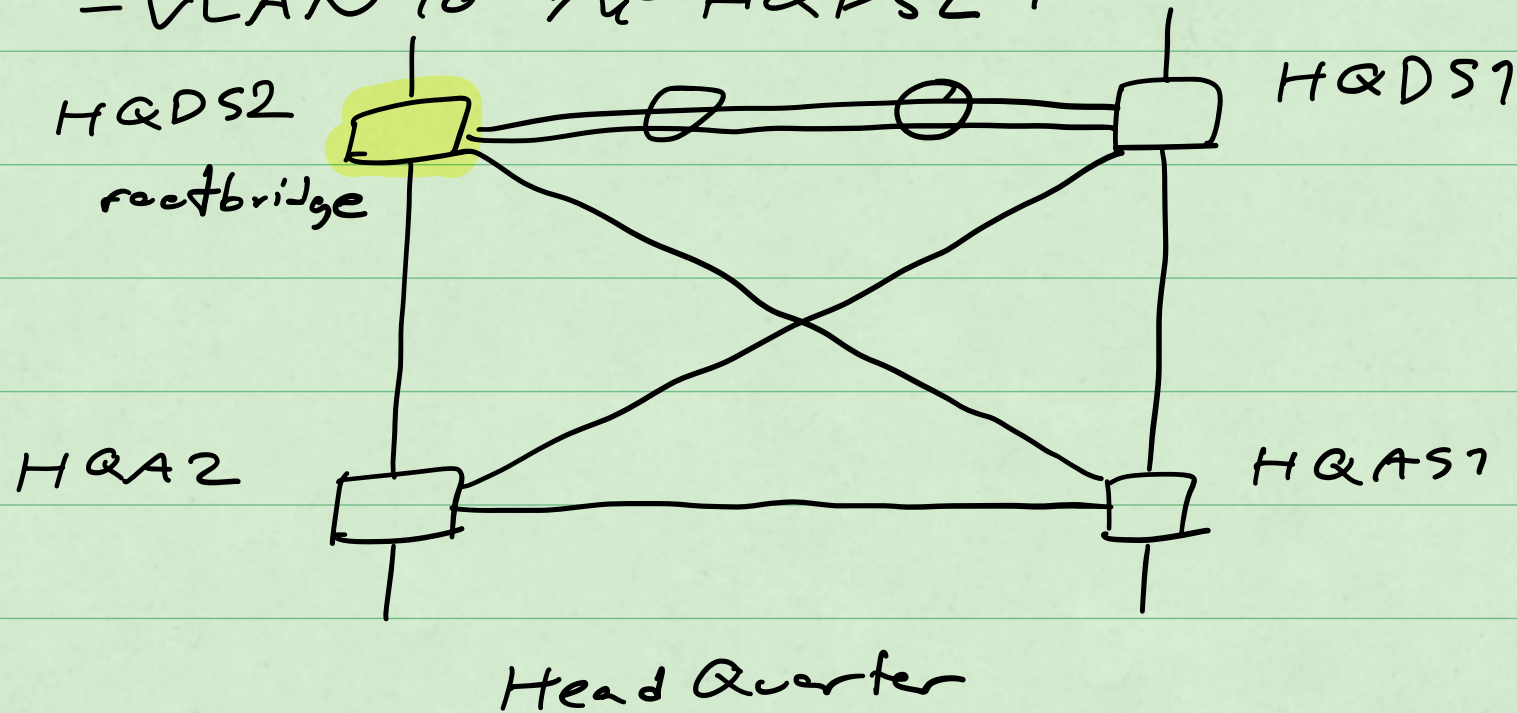
go/1.50 serve VLAN 50

RSTP and VLAN load balancing

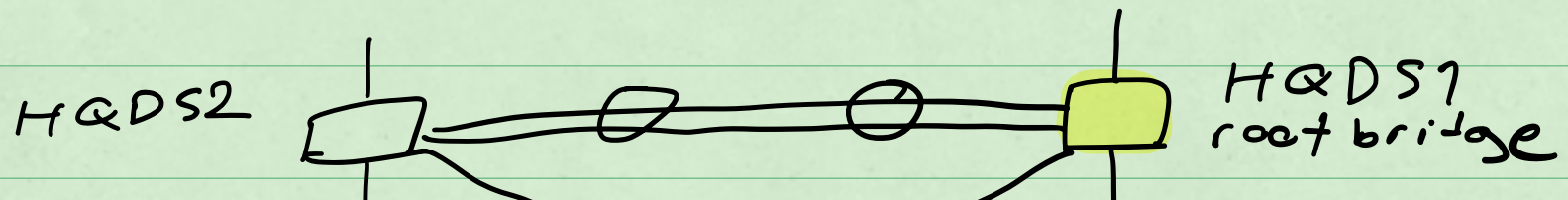
switch $\rightarrow$ no issue Head Quarter ID on RSTP

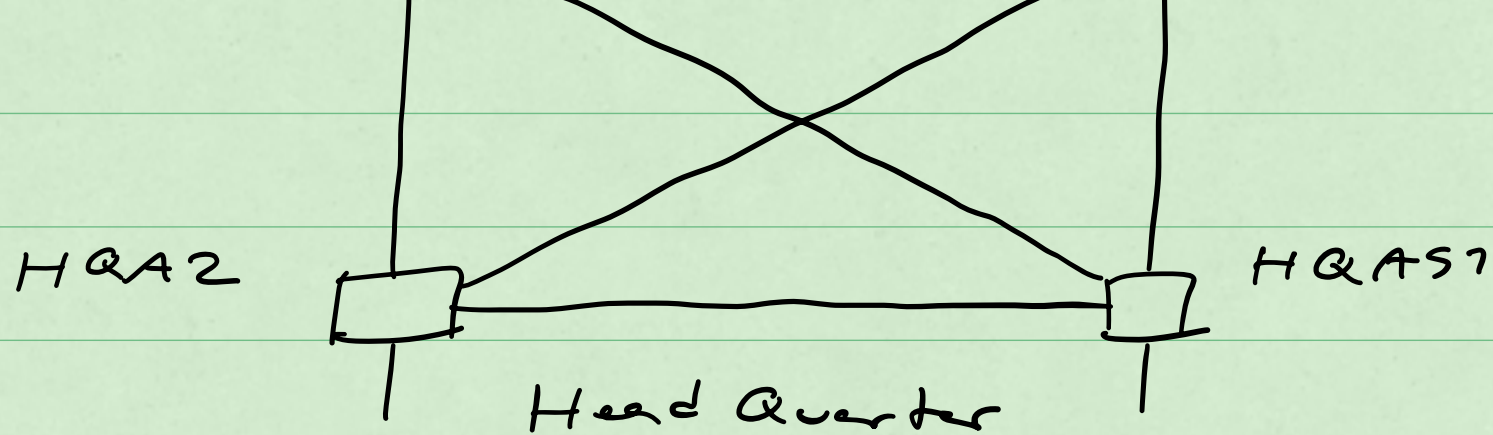
Load balancing

- VLAN 10 $\frac{1}{4}$ HQDS2 $\frac{1}{4}$ root bridge



- VLAN 20 $\frac{1}{4}$ HQDS1 $\frac{1}{4}$ root bridge





Options

access layer switch configuration port fast
 1102 bpdn guard running port 1102 for
 trunk port

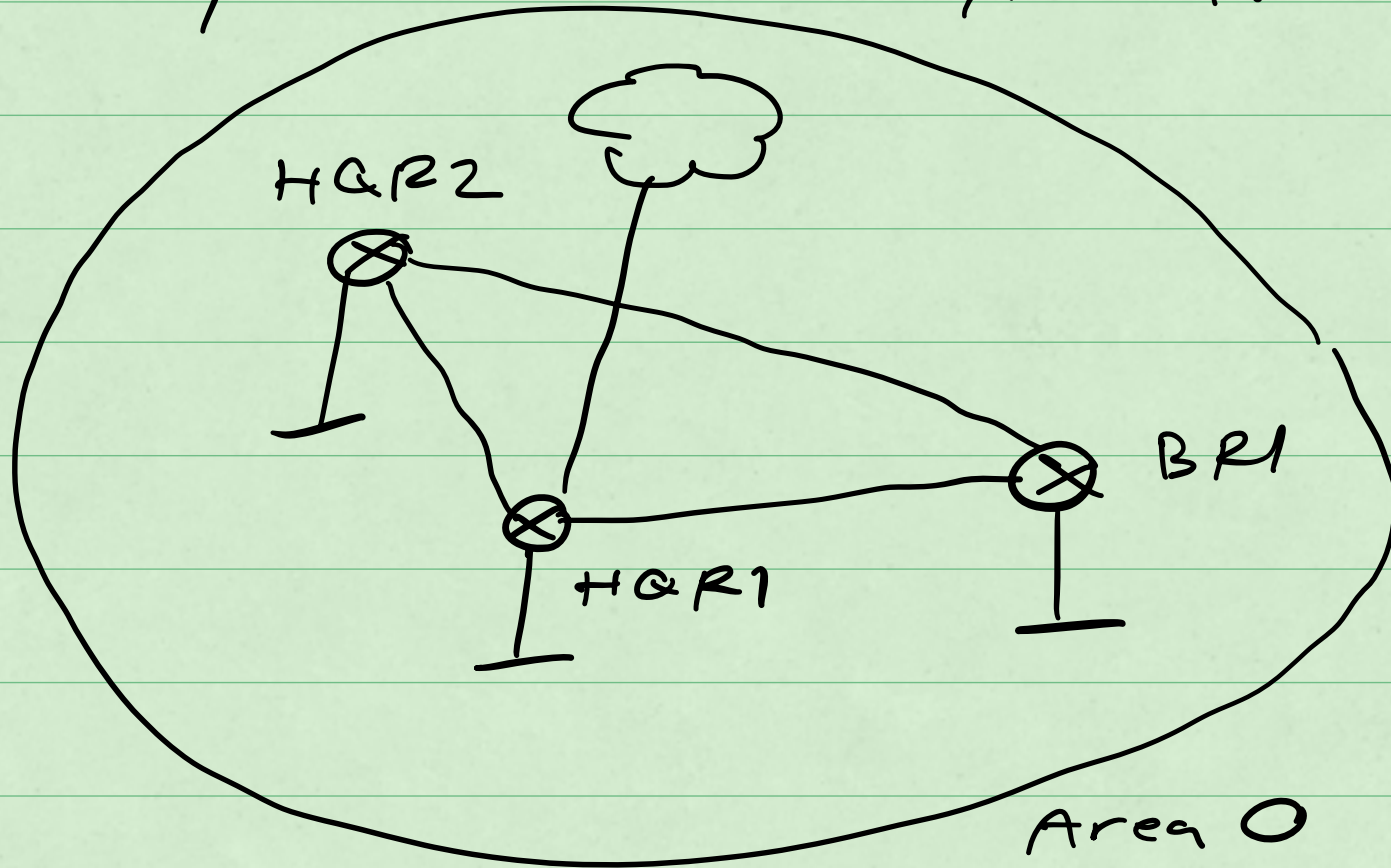
③ Layer 3

Static Routing

in HQA2 set default route on
 go to 192.0.2.1 internet

Dynamic Routing (OSPF)

router ከሰው ወይም area 0 ነው
በ-ነፃ passive interface ከሌለው በስተቀር



* LQR1 оцна default-information originate

FHRP (HSRP) and

in HSPP IDW = Head Center



VLAN 10 virtual IP no . .
 VLAN 20 virtual IP no

Load balancing

for vlan 10 HQR2 = active VLAN 10
 standby VLAN 20 VLAN 30

HQR1 = standby VLAN 10
 active VLAN 20 VLAN 30

④ Network Security & Access Control

Basic Device Security

Switch 110 = router 4110 = authentication
encrypt

username:

password:

Standard ACLS

policy ٧٥

7.

2.

3.

* $11011011 = 101$ standard ACLs 11101101 ICMP

מטעם שיש פארהאנדלע וואס זענען שטארק געווארן

۱۶۷۲ و ۱۶۷۳

Port Security

access switch ከባድ ነው MAC-address
sticky option ለውጦችን ጽድቅ VLAN 50
ገደብ፣ ሆኖም VLAN ጽንሰ-ሀሳብ

⑤ Network services DHCP

H&R2 ስለ BR1 ሆኖ DHCP server

- ስለ pool ሆኖ H&R2

excluded address — —

" " — —

:

ስለ ስለ pool ሆኖ pool ስለ ሆኖ VLAN

pool VLAN 10 — —

pool VLAN 20 — —

:

— monitor pool on BR1

o = exclude address — —
" " "

:

ip = monitor 2 pool also pool on 2 VLAN

pool VLAN 40 — —

pool VLAN 50 — —

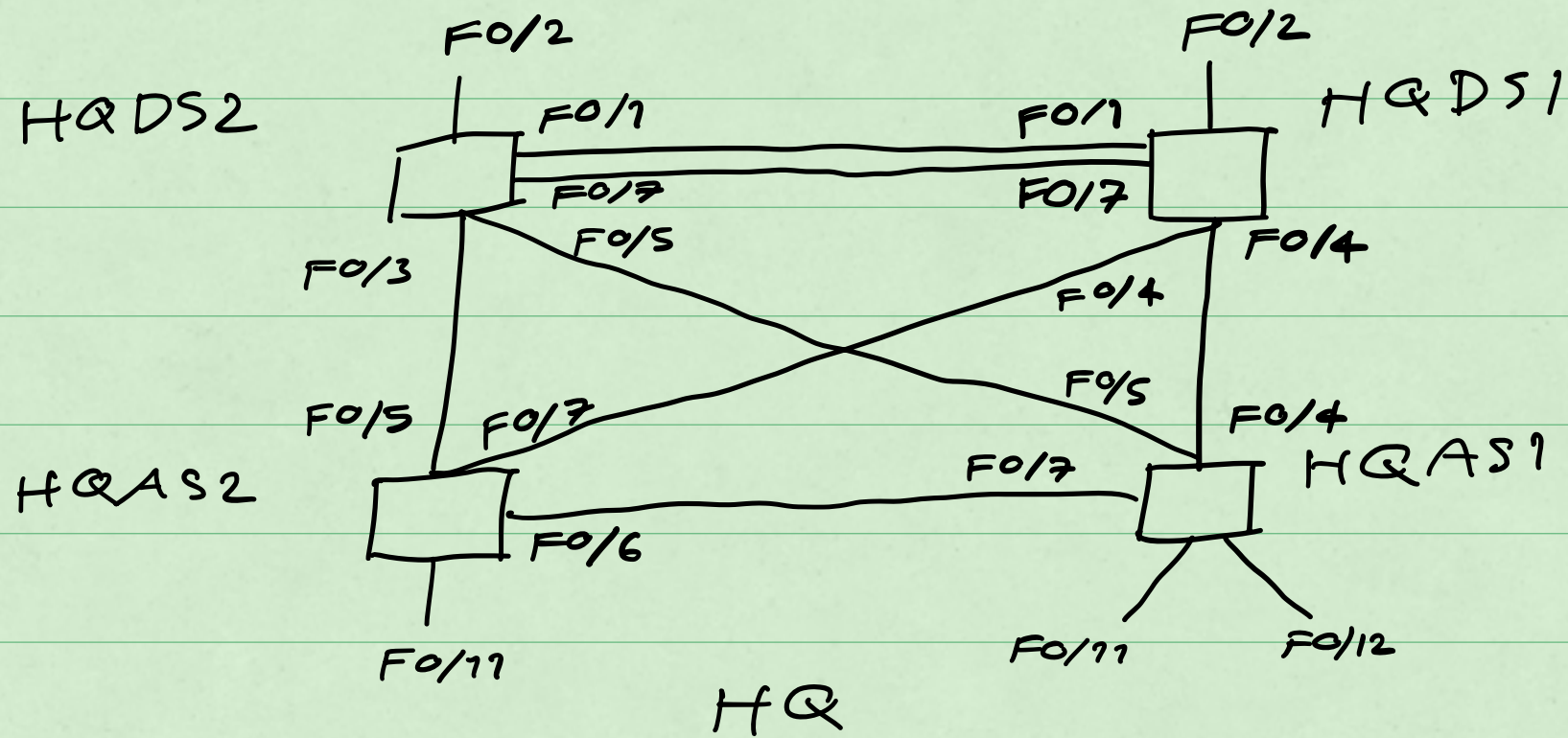
NAT Overload (PAT)

private IP

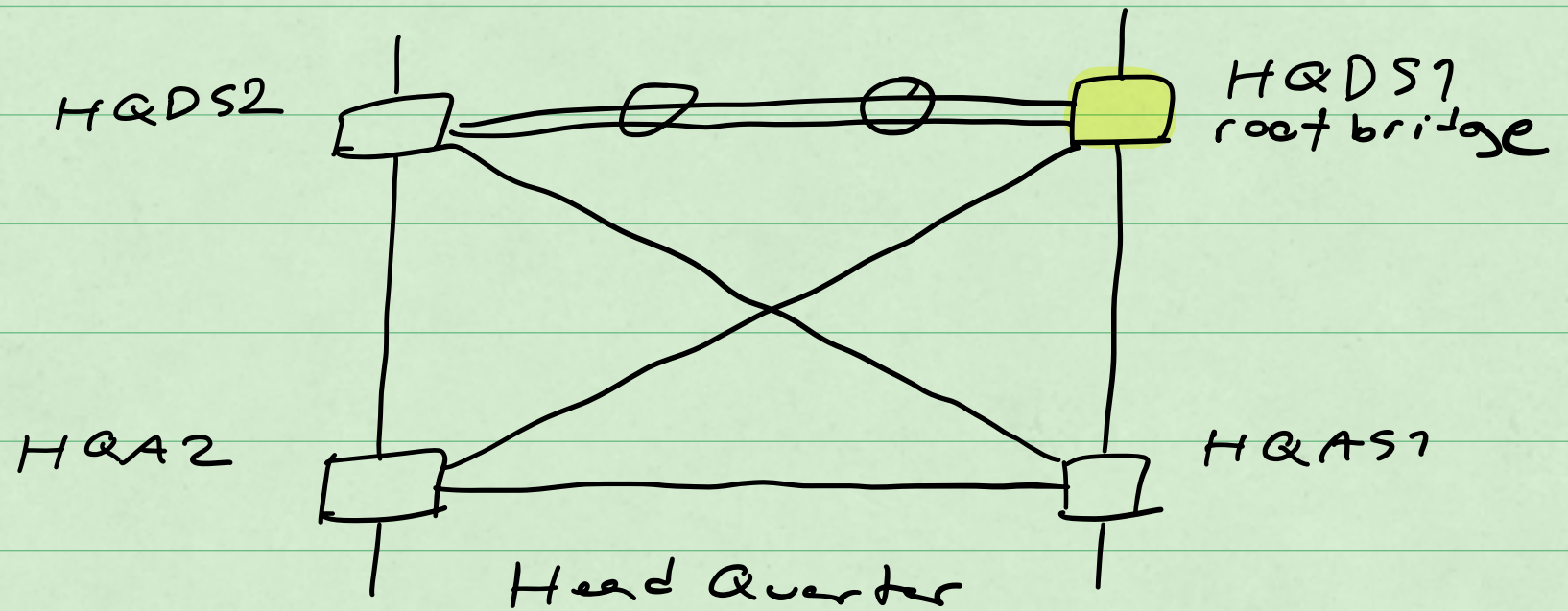
H&R1 o = 11.11.11.11 ← 172.16.0.0/24

o = 10.255.255.2/30 n = 100

port on transport layer

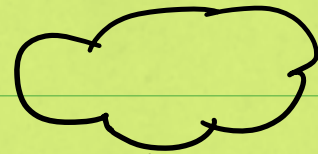


trunking



RSTP VLAN 20,30

AREA 0



INTERNET

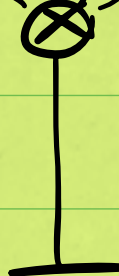
HQR2



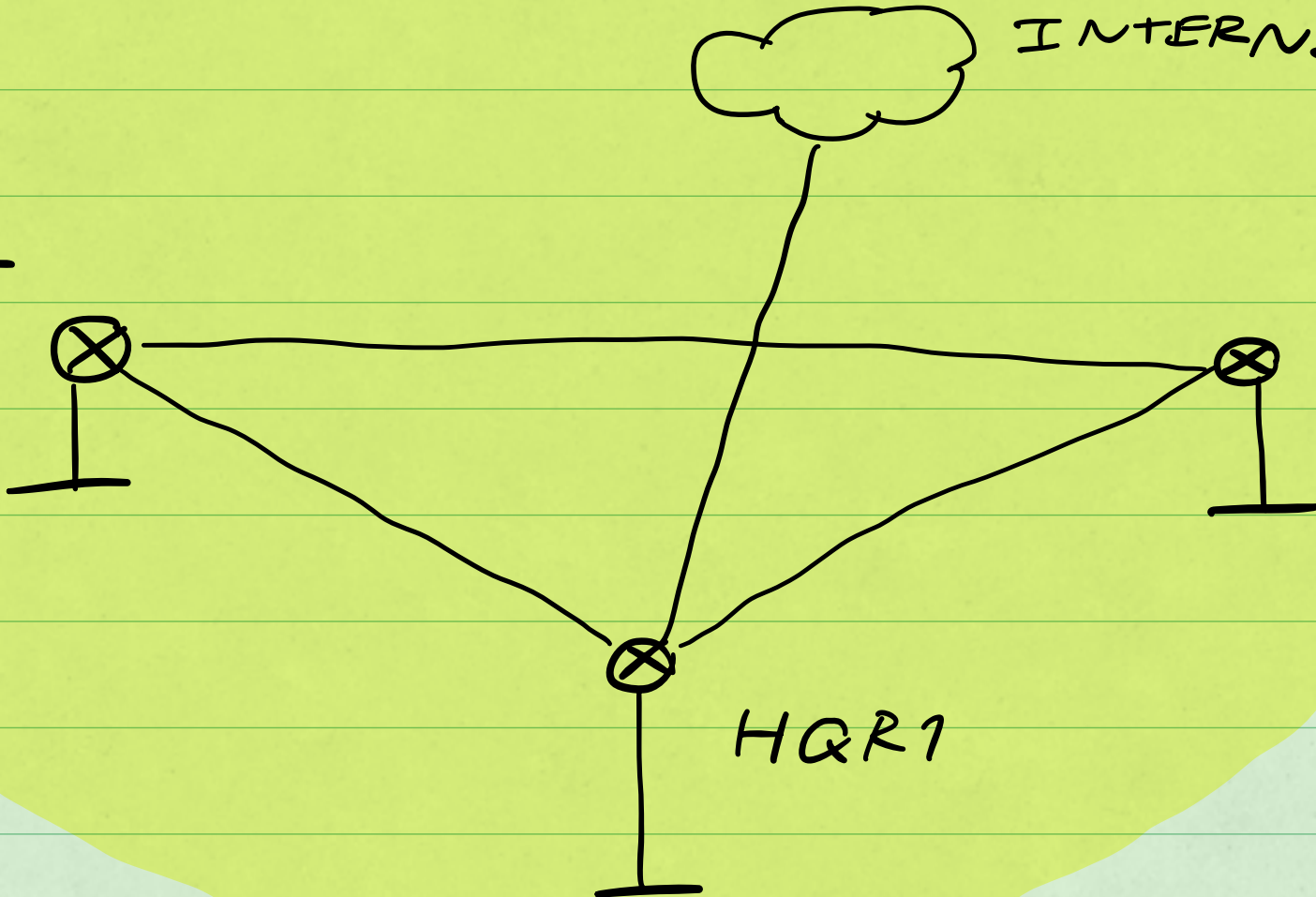
BR1

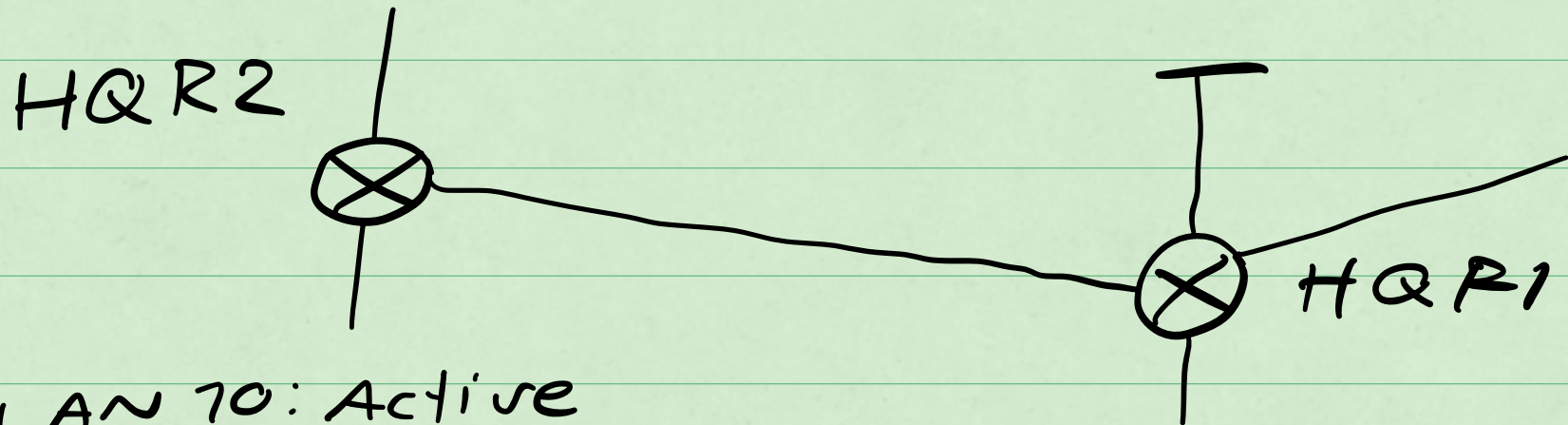


HQR1



OSPF





VLAN 70: Active
VLAN 20: Standby
VLAN 30: Standby

VLAN 70: Standby
VLAN 20: Active
VLAN 30: Active

HSRP